



Introduction to git - Part A & B

Installation & Setup



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1 | Goal

This lab is divided into two parts. Part A must be done at home as preparation, while Part B is done together in the lab. The lab will be done independently on your laptop and graded at the end.

In this lab we will learn the basic principles of version control [git \[1\]](#), in particular the tool [Sublime Merge](#) and optionally the [Git command line tool](#), which have to be installed and configured on your computer (see [Section 2](#)). In addition, an account is created on the [Github](#) platform.

Finally, we will learn the basics of Markdown in [Section 3](#) to easily write text files.



It is crucial that the installation and configuration is done carefully to avoid wasting time during part B of the laboratory.



In order for your course instructors to prepare the repositories on [Github](#), please provide your github username to them as soon as possible. How to do this is described in section [Section 2.3](#).



2 | Installation

The first step is to install Git and/or Sublime Merge. You can choose whether you want to use the command line or the GUI during the lab.

2.1 Sublime Merge

Visit the website <https://www.sublimemerge.com> then download and install the Sublime Merge tool [2].

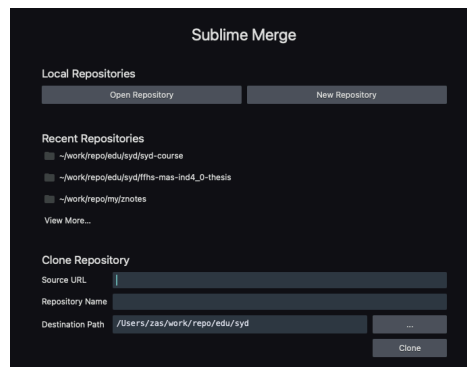


Figure 1 - Sublime Merge GUI

2.1.1 Configuration

When cloning a repository, Sublime Merge will automatically ask for your identity and prompt you to log in to your Github account.



2.1.2 Overview

The interface of Sublime Merge is presented in the [Figure 2](#):

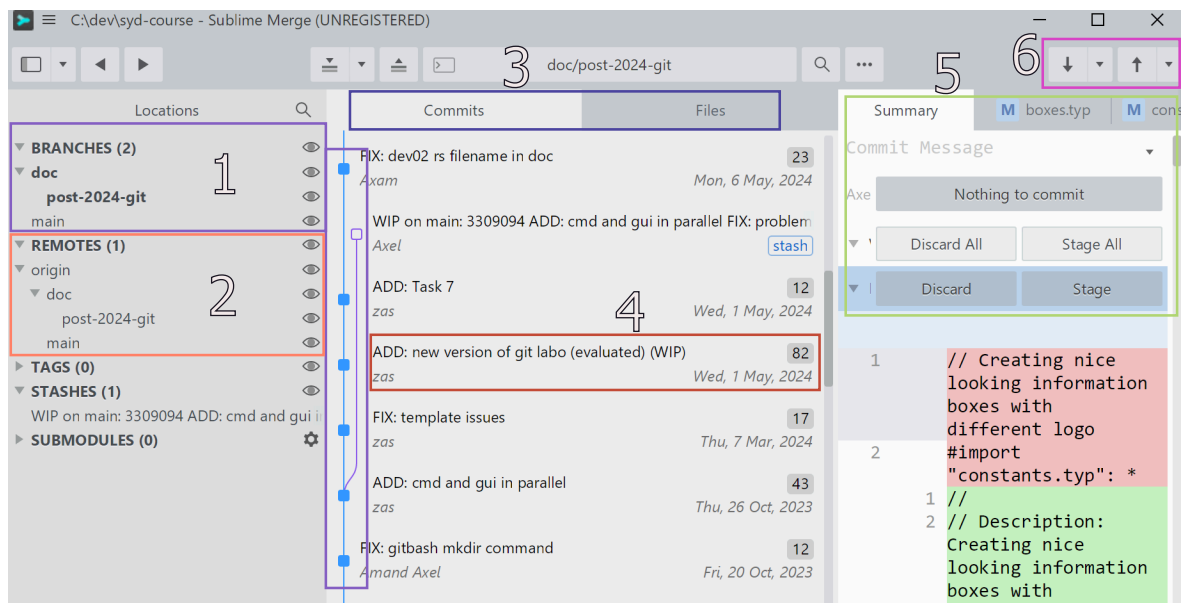


Figure 2 - Sublime Merge GUI

1. **Branches**: list of branches of the repository. You can create, delete and rename branches. They are also displayed in their timeline under the **Commits** tab (point 3).
2. **Remotes**: list of remote repositories, i.e. the servers on which the code is stored and to which you can **push**.
3. **Commits/Files**: tabs to switch between the chronological view of commits and the view of changes for the selected commit.
4. **Commit description**: a commit consists of a message, an author and a date.
5. **Current changes**: the files that have been modified compared to the last commit are listed here. You can select the files you want to **commit** by selecting them and clicking on the **Stage** button. As long as the files have not been **committed**, you can also remove a file from staging by clicking on the **Unstage** button or completely delete a file's changes by pressing **Discard** (⚠ permanent deletion ⚠).
6. **Pull / Push**: the two buttons allow you to **pull** - take the latest changes from the remote repository - and **push** - send local changes to the remote repository. It is also possible to **fetch** by clicking on the small arrow next to the **Pull** button, which allows you to see the new commits without modifying the local repository.



2.2 git - command line

You can download the latest version from the official website <https://git-scm.com/> [1]. Git is available for Linux, Mac, and Windows. This lab requires git ≥ 2.27.

Start “Git Bash” on Windows or “Terminal” on MacOS. This is a Unix/Linux-like command editor that allows you to run Git commands in console mode.

```

Last login: Tue Mar  8 09:26:26 on tty004
[~] zsh xac [~] (base)
[~] % git --help
usage: git [--version] [--help] [-C <path>] [-c <name>=<value>]
      [--exec-path<gitpath>] [--html-path] [--man-path] [--info-path]
      [-p | --paginate] [-P | --no-pager] [--no-replace-objects] [--bare]
      [--git-dir<gitpath>] [--work-tree<gitpath>] [--namespace<name>]
      [--super-prefix<path>] [--config-env<name>=<envvar>]
      <command> [<args>]

These are common Git commands used in various situations:

start a working area (see also: git help tutorial)
  clone Clone a repository into a new directory
  init Create an empty Git repository or reinitialize an existing one

work on the current change (see also: git help everyday)
  add Add file contents to the index
  mv Move or rename a file, a directory, or a symlink
  restore Restore working tree files
  rm Remove files from the working tree and from the index

examine the history and state (see also: git help revisions)
  bisect Use binary search to find the commit that introduced a bug
  diff Show changes between commits, commit and working tree, etc
  grep Print lines matching a pattern
  log Show commit logs
  show Show various types of objects
  status Show the working tree status

grow, mark and tweak your common history
  branch List, create, or delete branches
  commit Record changes to the repository
  merge Join two or more development histories together
  rebase Reapply commits on top of another base tip
  reset Reset current HEAD to the specified state
  switch Switch branches
  tag Create, list, delete or verify a tag object signed with GPG

collaborate (see also: git help workflows)
  fetch Download objects and refs from another repository
  pull Fetch from and integrate with another repository or a local branch
  push Update remote refs along with associated objects

'git help -a' and 'git help -p' list available subcommands and some
concept guides. See 'git help commands' or 'git help concepts'
to read about a specific subcommand or concept.
See 'git help git' for an overview of the system.

[~] % git xac [~] (base)
[~] %
  
```

Figure 3 - git Terminal



Note that for all commands in Git Bash, you can get help by inserting `--help` after the command.

```
git --help
```

2.2.1 Global configuration

A variety of settings can be configured in Git. It is possible to change the settings globally on your computer (flag `--global`) or only for a specific repository.

We will now perform the minimal configuration. Use the following commands to set your identity in Git globally on the system. Use your name and email address. This information is publicly visible to identify your work (your commits).

```
git config --global user.name "Firstname Lastname"
git config --global user.email first.last@email.ch
```

For example:

```
git config --global user.name "Silvan Zahno"
git config --global user.email silvan.zahno@hevs.ch
```

You can check the configuration with the following command:



```
git config --list
```

You can also check a specific setting:

```
git config user.name
```

2.3 Online accounts

2.3.1 Github

In the laboratory, you will be working with a Git repository on [Github](https://github.com). You will need a login to do this. If you haven't created an account yet, you will need to create one. If you already have one, you can use that.

2.3.1.1 Create a New Account

Visit the website <https://github.com> then create an account and log in.

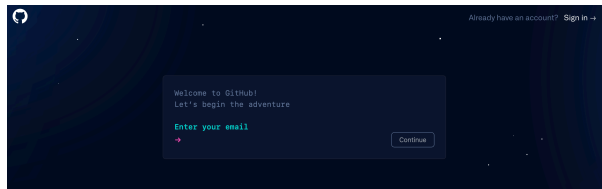


Figure 4 - GitHub Login

2.3.1.2 Provide Username to Instructors

You need to provide your github username to the professor or teaching assistant in order they can set up the git repository and invite you to collaborate.

In case you do not know your github username you can find it as follows:

- Log in to the [Github](https://github.com) website
- Click on your profile picture on the upper right corner and select **Profile**:

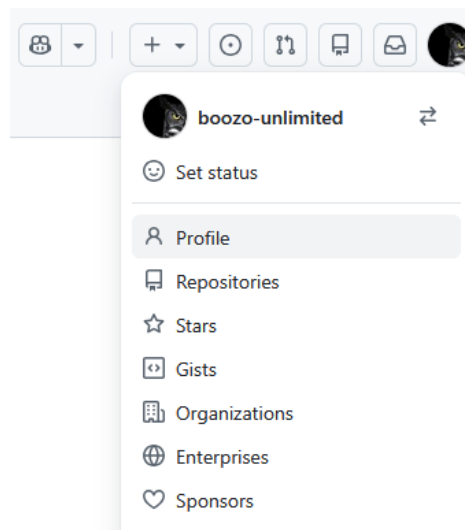


Figure 5 - GitHub Profile



- Find your username below your profile picture:

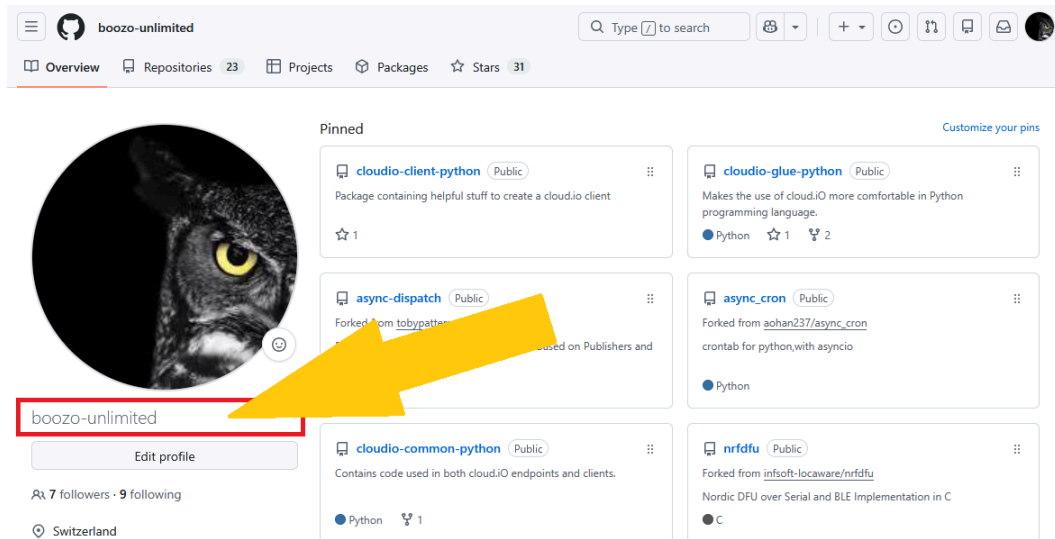


Figure 6 - GitHub Username Location

- Copy the username and provide it to one of your instructors. Preferably to [Thomas Sterren via Teams](#).

2.4 Windows Configuration

In order to see also the hidden `.git/` folder as well as file extensions, configure your Windows File Explorer as follows:

File Explorer ⇒ View ⇒ Show ⇒ Activate **“File name extensions”** and **“Hidden items”**

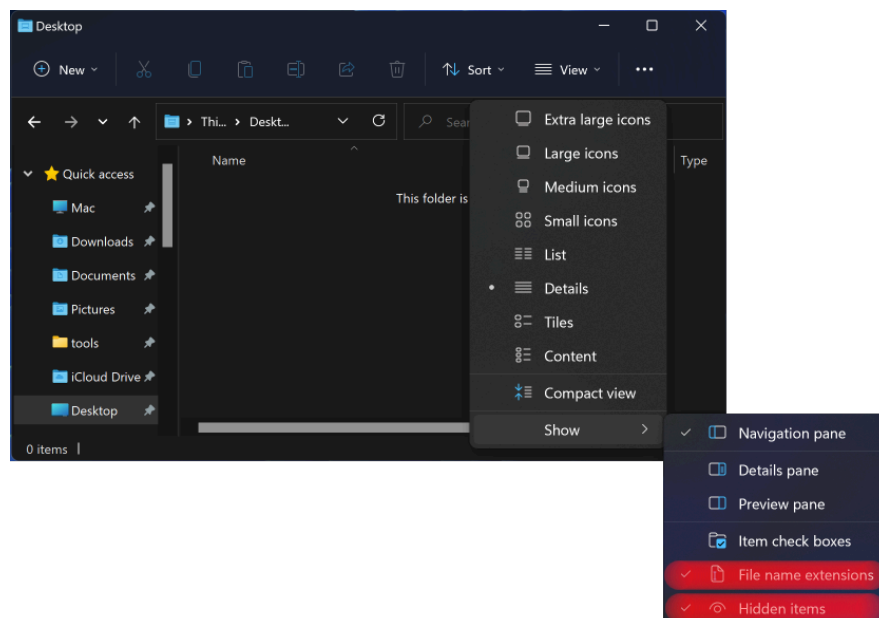


Figure 7 - Windows File Explorer Configuration



3 | Markdown

Markdown is a lightweight markup language with plain-text formatting syntax. It is designed to be easy to read and write, while also being easily converted into PDF, HTML or other formats. Markdown is commonly used for formatting text on the web, such as in README.md files, documentation, forum posts, and messaging.

In order to write Markdown, you need your preferred Text editor supporting the format:

- **Sublimetext** with the **MarkdownLivePreview** plugin
- **Zed**: open markdown file and use Ctrl/Cmd + Shift + V to open the preview
- **VSCode**: open markdown file and use Ctrl/Cmd + Shift + P => Markdown: Open Preview to the Side
- **Markdown Live Preview**: online markdown editor
- ...

For example the [intropage](#) of this course is written in Markdown, you can see the source code of the page by clicking on the “Edit this page” button on the top right corner of the page or via this [link](https://raw.githubusercontent.com/hei-synd-syd/.github/main/profile/README.md) - <https://raw.githubusercontent.com/hei-synd-syd/.github/main/profile/README.md>.

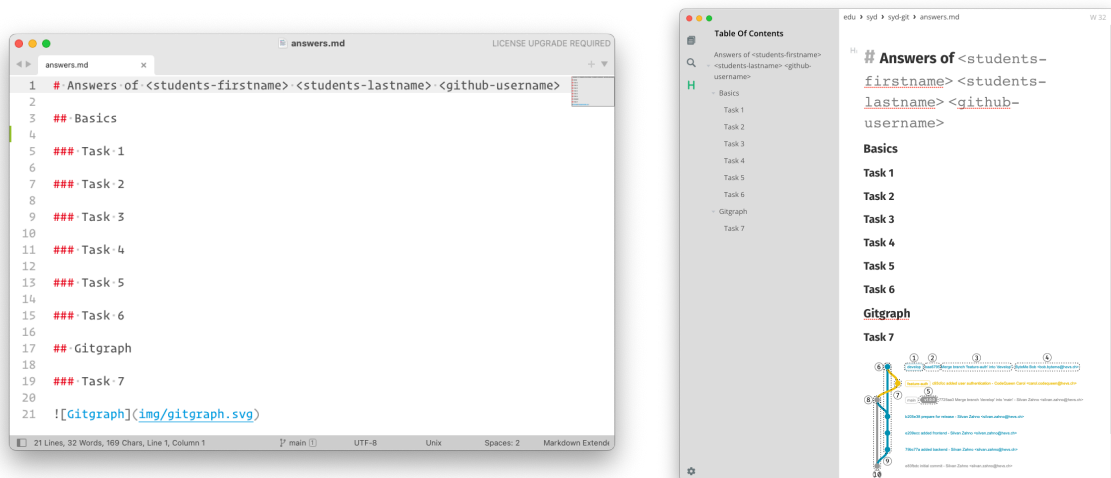


Table 1 - Left: Sublime Text, Right: Marktext



For the lab, you will need to write a documents in Markdown format. Be ready with your editor.



3.1 Markdown syntax

Hereafter a short overview about how a markdown file is structured. The syntax is simple and easy to learn. The file has to be saved with the extension `.md`. A more complete syntax list can be found at https://wiki.zahno.dev/multimedia/writing/md/md_github.html.

```
# Title 1

## Title 2

### Title 3

Some simple Text italic bold
Strikethrough `monospaced`

Fomulas 
$$S = \sum_{i=1}^n x_i^2$$


- List Item 1
- List Item 2
1. Numbered List Item 1
2. Numbered List Item 2

[Link name](https://hevs.ch/synd)

![logo](logo.svg)

```rust
// A rust code bloc
fn main(){
 println!("Hello World");
}
```

Tables	Are	Cool
col 1	left-align	$f_{clk}$
col 2	centered	$12
col 3	right-align	1024
```

urces > git > markdown-cheatsheet.md W 90

Title 1

Title 2

Title 3

Some simple Text *italic* **bold** ~~Strikethrough~~ `monospaced`

Fomulas
$$S = \sum_{i=1}^n x_i^2$$

- List Item 1
- List Item 2

- Numbered List Item 1
- Numbered List Item 2

[Link name](#)



```
// A rust code bloc
fn main(){
    println!("Hello World");
}
```

| Tables | Are | Cool |
|--------|-------------|-----------|
| col 1 | left-align | f_{clk} |
| col 2 | centered | \$12 |
| col 3 | right-align | 1024 |



4 | Outro

Congratulations You have now installed and configured everything you need to work with Git. You should have:

- ☐ Installed Git and Sublime Merge
- ☐ Configured git with your name and email
- ☐ Created a GitHub account
- ☐ Familiarized yourself with the Sublime Merge graphical user interface (GUI)
- ☐ Familiarized yourself with the Git theory and commands
- ☐ Familiarized yourself with Markdown and its syntax and be able to write and preview Markdown documents



See the appendix [Section A](#) and [Section B](#) for a summary of the most important Git commands.



5 | Goals

In this lab we will learn the basic principles of version control [git \[1\]](#). You must have already done part A of the lab at home to be able to start with part B.

In [Section 6](#) we learn the basic operations to be able to work with Git. The created repository will then be published on [GitHub](#). The advanced functions [branch](#) and [merge](#) are tried out in an example in [Section 7](#). In [Section 9](#) we all work together on a repository. Finally, there is some optional work in [Section 10](#).



At the end of the lab, make sure you have published all changes on GitHub. Only the changes published on GitHub will be evaluated.

5.1 Tools

In this lab you will use **Sublime Merge** as a graphical tool or **Git CMD** as a command line.



Git CMD
App

Figure 8 - Git CMD Commandline



Sublime Merge
App

Figure 9 - Sublime Merge GUI



Use the command line process only if you are comfortable with a terminal. You will need to know the commands to navigate, display information, etc. `cd`, `ls`, `cat`, `touch`, `nano` ...



5.1.1 Overview

The interface of Sublime Merge is presented in the [Figure 10](#):

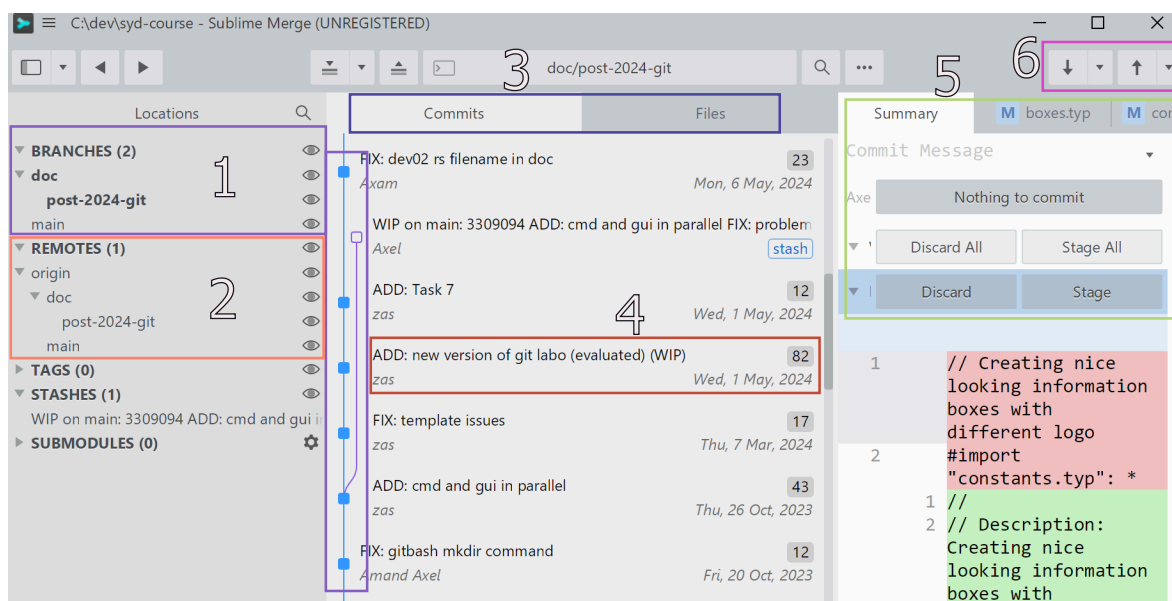


Figure 10 - Sublime Merge GUI

1. **Branches**: list of branches of the repository. You can create, delete and rename branches. They are also displayed in their timeline under the **Commits** tab (point 3).
2. **Remotes**: list of remote repositories, i.e. the servers on which the code is stored and to which you can **push**.
3. **Commits/Files**: tabs to switch between the chronological view of commits and the view of changes for the selected commit.
4. **Commit description**: a commit consists of a message, an author and a date.
5. **Current changes**: the files that have been modified compared to the last commit are listed here. You can select the files you want to **commit** by selecting them and clicking on the **Stage** button. As long as the files have not been **committed**, you can also remove a file from staging by clicking on the **Unstage** button or completely delete a file's changes by pressing **Discard** (⚠ permanent deletion ⚠).
6. **Pull / Push**: the two buttons allow you to **pull** - take the latest changes from the remote repository - and **push** - send local changes to the remote repository. It is also possible to **fetch** by clicking on the small arrow next to the **Pull** button, which allows you to see the new commits without modifying the local repository.



6 Basic Operations



The answers to the questions **must** be written down in the Markdown file `answer.md`. The file is located in the repository that will be mentioned in the next step.

6.1 Creating a git repository

In order to work on a common basis, we will work with a copy of the template repository. This operation is called `fork` - [Figure 11](#). This fork creates a full copy of the original repository which is hosted somewhere else on Github. The changes will then be reflected in the newly created repository, not on the original one. You can therefore work on it without disturbing the work of others.

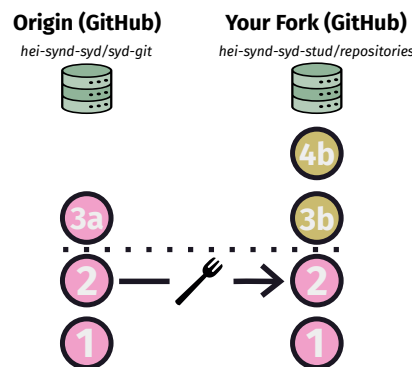


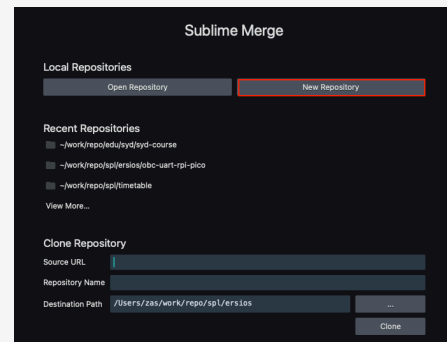
Figure 11 - Fork of the template repository



If you want to create a new repository from scratch (not needed in this labo), you can do so with the following commands or via Sublime Merge GUI:

```
cd /path/to/my/repo  
git init my-new-repo
```

File ⇒ New Repository



The newly created repository will be empty, without a remote which means without a connection to a remote repository (e.g. GitHub). You can add a remote repository later with the command `git remote add origin <url>` or via the GUI.

Over the last few days, your instructor has created a repository for you, which you will be working on. You have received an email invitation asking you to join it. Once you have accepted the invitation, you will gain access to your repository.

The repository is located in the following organisation: [GitHub hei-synd-syd-stud](#)



If you have not received an invitation from your instructor or cannot find a repository via the link provided above, please contact your instructor!

Please use this repository for the following tasks.



Please ensure that you are working with the repository provided (and not with a fork of your repository, etc.). Only the results in the provided repository will be taken into account for evaluation!

6.2 Clone

Go to the [hei-synd-syd-stud](#) page in your repository and copy the link, which you can then use to create a clone on your computer:

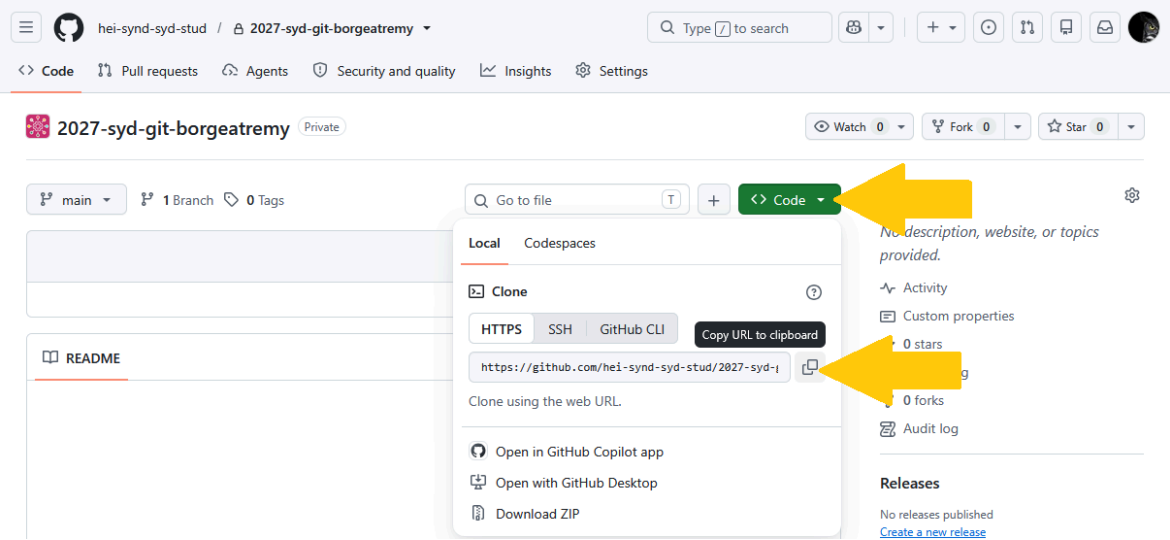


Figure 13 - Example link to Clone your Repository

Clone it to your local computer at a location of your choice.

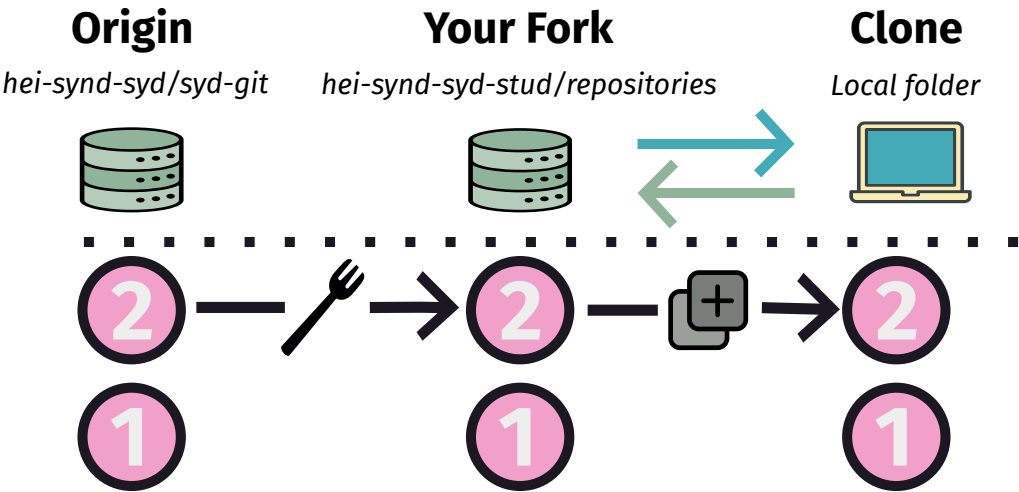


Figure 14 - Cloning

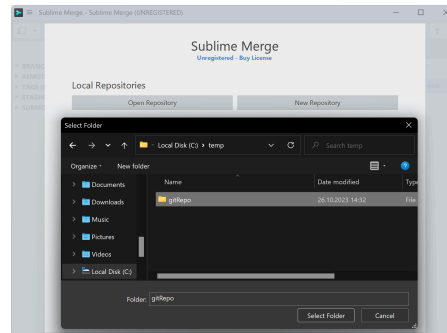


Commandline

```
cd /path/to/my/repo
git clone <linkurl>.git

# Example
git clone https://github.com/hei-synd-syd-stud/
2025-syd-git-tschainz.git
```

GUI - CTRL+T ⇒ Paste Source URL ⇒ Select Folder



6.3 Get status

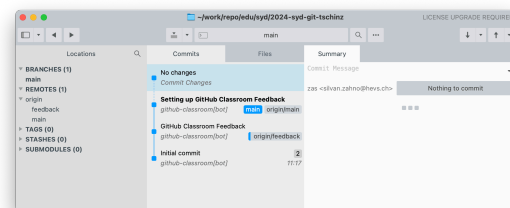
Get information about your repo:

Commandline

```
git status
git log --oneline
```

GUI

See the status in the main window.



Currently, your repository has no changes since nothing has been modified. In Sublime Merge, at the top of the window, it says No changes. On the right side, you see Nothing to commit meaning we have a copy of the remote repository with no changes added to it.



6.4 Stage & commit

We will now modify data in the repository, in this case the file `answers.md`.

Task 0

Open the document `answers.md` and replace:

- `[students-firstname]` with your first name
- `[students-lastname]` with your last name
- `[github-username]` with your GitHub username



REMOVE THE “[]” from your names.

In the following sections, the **Exercises** must be answered in this same file `answers.md`.

Always **SAVE** your file after answering a task.

Like shown under [Section 6.3](#), check the status of your repo.

Task 1



What is the status of the repository / what do you see different? What does it mean?

We are going to save the changes locally. To do this, it is necessary to **Commit** the changes.

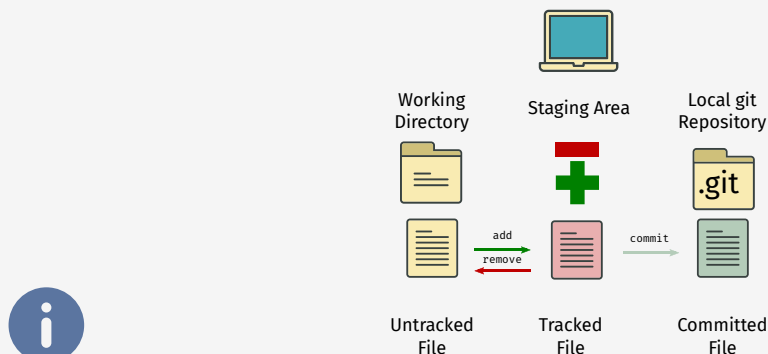


Figure 15 - Types of local Git operations

Your local git repo consists of three areas that are maintained by git:

- Working directory is a directory that contains the current version of your files (a normal file directory in the eyes of your operating system).
- Stage contains the changes to be included in the next commit.
- Head points to the location in the Git repo tree where the next commit should be made.



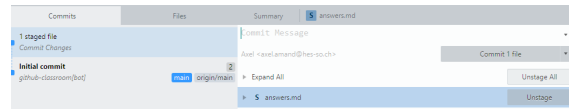
To save the changes *locally*, you first need to select the changes that will be saved in the repo. This process is called **stage**.

Commandline

```
git add answers.md
```

GUI

Select the file in the upper-right corner, and click on stage.



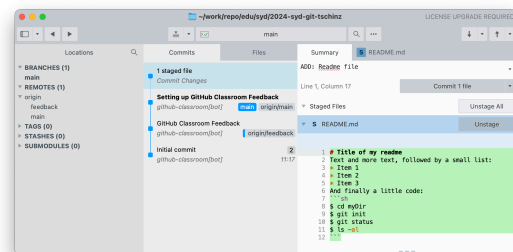
Once all changes are selected, the save is done through a **Commit**. A commit is a snapshot of your repo at a given time and once done, the state of the folder is saved in the repository. The **Commit** is identified by a unique identifier (hash) and contains information about the author, date and commit message, which should reflect the changes made:

Commandline

```
git commit -m "CHG: personal data in  
answers.md"
```

GUI

Commit Message ⇨ CHG: personal data in
answers.md ⇨ Commit 1 file



Commit your changes

Task 2

After you have made a commit as explained above, answer the following questions:



- Why is the example commit message worded this way: CHG: personal data in answers.md ?
- Is it possible to create a commit without a message?
- What is the status of the repo now?
- Is the remote repo (here Github) up to date with our changes?



6.5 Add files

Then create an empty file named `README.md` in the main directory of your repository:

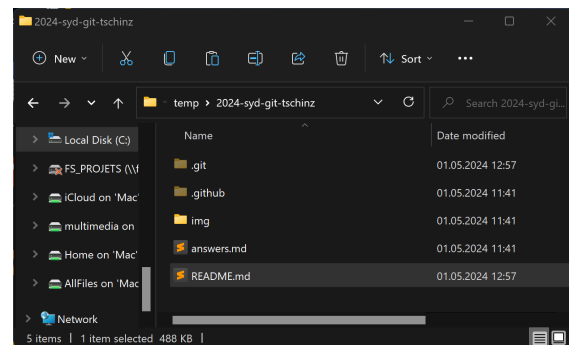
Commandline

```
touch README.md
```

GUI

(Windows) `C:\path\to\repo\` ⇒ New ⇒ Any File
⇒ `README.md`

(Linux&MacOS) `/path/to/repo` ⇒ New ⇒ Any File
⇒ `README.md`



Task 3

Have another look at the status of your repo, what do you see?



Make a commit with the new file and the answers to exercises 2 and 3.

Example repository

A simple Git repo consisting of five commits can be represented as follows. The `Head` position is a reference to a commit that represents the current state/view of the repo, in this case the latest change:

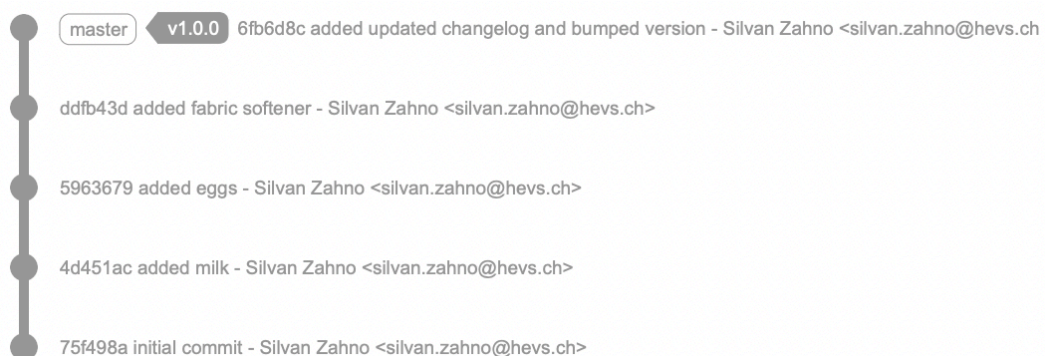


Figure 16 - Five commits on the local repo, each commit has its own identification



6.6 Markdown

Modify the README.md file with a text editor to make it resemble roughly like the following:

My README.md

The goal of this document is to :

- Approach the Markdown syntax
- Show the result on a Github repository

How-To

1. Identify constructs from given result
2. Write a corresponding Markdown syntax

Git - a command example

Git can be ran through command lines and the current status shown with:

```
# Going inside directory
cd /my/git/dir
# Asking for current status
git status
git log --oneline
```

It is also important to note that:

"It is easy to shoot your foot off with git, but also easy to revert to a previous foot and merge it with your current leg."

--- Jack William Bell

Figure 17 - Markdown structure to reproduce

Use a tool as shown in chapter Labo Part A - Markdown to help you write and preview your markdown file.

Here is a Markdown cheatsheet reminder to help you create your README.md file:

```
# Title
## Subtitle
### Sub-subtitle
```

Some text

Some code:

```
```bash
cd myDir
```
```

A list:

- * Item 1
- * Item 2

> Quoting something

A numbered list:

1. Item 1
1. Item 2



Complete your README.md file.



Once satisfied, commit your changes.



6.7 Push online

Currently, all the work done is only saved *locally*. To put it online, it is necessary to push the commits to the remote repository (Github) in order to have a copy of it and, as part of group work, allow others to see it:

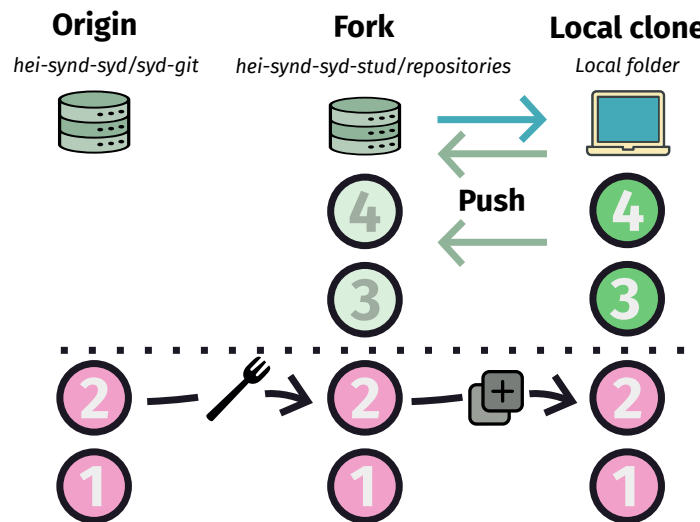


Figure 18 - Push to the remote repository

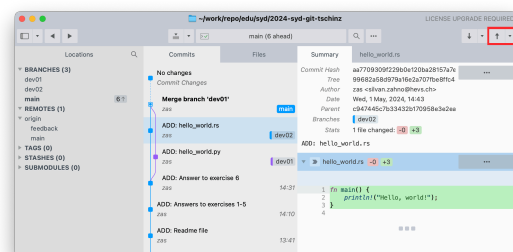
To do this, once all changes have been committed:

Commandline

```
git push origin main
```

GUI

Top Right Arrow ⇒ Push changes



Go online to Github to see the result of your work.

You should see the `README.md` file with the content you created in readable form.



6.8 Back in time (not the future)

Git is a version control system. It is therefore possible to return to an earlier version of the repository. To do this, you first need to identify the commit you want to go back to. Here we go back to the very first commit.

Every commit is provided with a “hash” or “checksum” (of type sha1). This hash is a unique identifier that allows you to isolate a specific commit. The latter is visible by clicking on a commit for SublimeMerge, or by using the following command:

```
git log --oneline
```

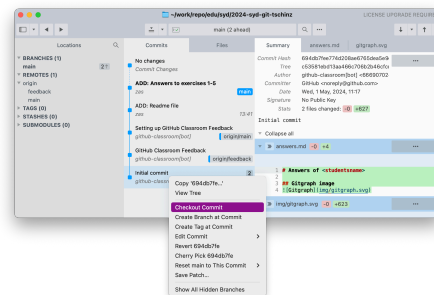
To change commits, perform the checkout operation:

Commandline

```
git checkout <SHA1>
# for example
git checkout 694db7f
```

GUI

Select First Commit ⇒ Right click ⇒ Checkout commit



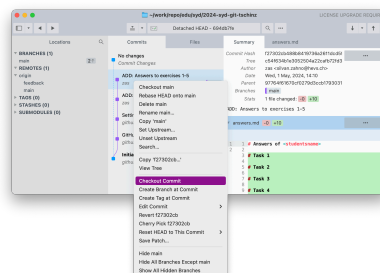
Then go back to the last commit in the same way. Apart from the hash, it is possible to return to the last commit by using the name of the branch, here main.

Commandline

```
git checkout main
```

GUI

Branches (1) ⇒ main ⇒ checkout main



Task 4

What happens in your working directory when going back to the “Initial commit” commit? What happens when going back to the last commit?



6.9 General questions



Task 5

What is the difference between the local repository and the remote repository?

What would happen if you deleted the local repository?



Task 6

What about the [original repository](#) that was forked?

Has it been modified?



Make a commit containing the previous answers.



Push your changes to the remote repository on Github before continuing.

6.10 Tag

To identify key commits, it is possible to create `tags`. A tag is a pointer to a particular commit. It is often used to mark versions of code, for example `v1.0`, `v2.0`, etc.

To mark the end of this chapter, create a tag `chapter2` on the last commit. To do this:

Commandline

```
git tag chapter2
git push origin chapter2
```

GUI

- Right click on commit ⇒ Create Tag at Commit ⇒ chapter2 ⇒ Enter
- Right click on tag ⇒ Tag chapter2 ⇒ Push Tag chapter2



Check online on Github that the tag `chapter2` has been successfully created and pushed, along with your latest changes.



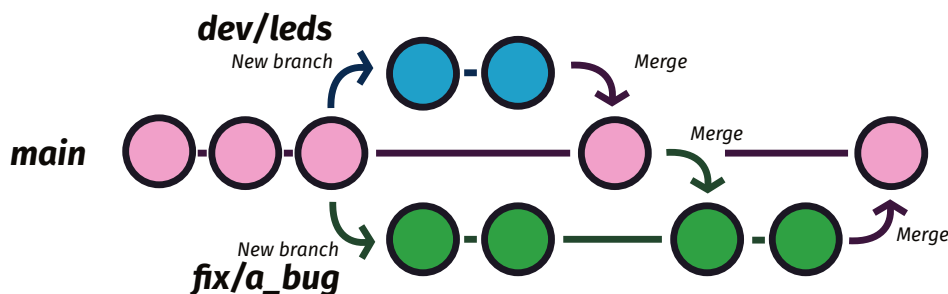
This chapter is to be completed in the same repository as before.



Make sure you have committed and pushed all your changes before starting this chapter, and that these have been tagged by chapter2.

7 | Branch and Merge

So far, we have used the basic functions of git - **pink commit line**. There are also the **branches** and **merge** functions, which Git has greatly simplified compared to previous tools:



Branches allow you to work in parallel on multiple versions of the same project. For example, you can create a **branch to develop a new feature**, while your colleague works on a **patch to fix a bug**. The merge action (**merge**) allows you to combine changes made on one branch into another.



Git is not omniscient. If two branches have modified the same files, Git will not know how to merge them, and you will have to manually resolve the conflicts.

7.1 First Branch

The first branch simulates the work of a first person on a new file, `hello_world.py`. This could be any file: code, image, Inventor drawing, Matlab simulation ...

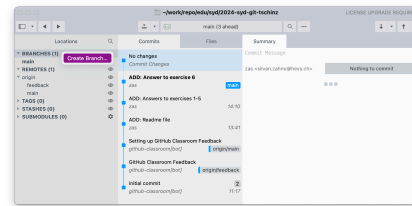
1. Create a development branch `dev01` in your local repo:

Commandline

```
git checkout -b dev01
```

GUI

Branches ⇒ Create Branch ⇒ dev01



2. Create a commit on this branch:

- Create a file `hello_world.py`
- Fill it with the following code:

```
print("Hello, world!")
```



Make a commit as presented in chapter [Section 6.4](#), with a clear message.

7.2 Second Branch

The second branch simulates the work of a second person on another file, `hello_world.rs`.

Assuming the branch was created at the same time as the other, it is necessary to first return to the same commit as before.

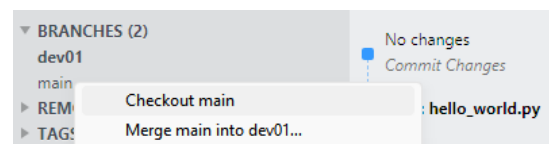
1. Switch back to the `main` branch:

Commandline

```
git checkout main
```

GUI

`main` ⇒ Right click ⇒ Checkout main



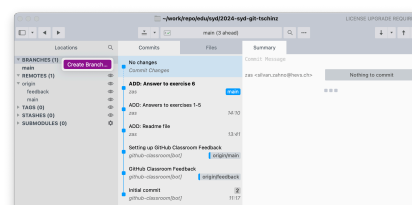
2. From the `main` branch, create a new development branch `dev02`:

Commandline

```
git checkout -b dev02
```

GUI

`Branches` ⇒ `Create Branch` ⇒ `dev02`



3. Create a commit on this branch:

- Create a file `hello_world.rs`
- Fill it with the following code:



```
fn main() {  
    println!("Hello, world!");  
}
```



Make a commit as presented in chapter [Section 6.4](#), with a clear message.



7.3 Merge dev02

Now that both branches have received commits, it is time to merge them into the main branch `main`.

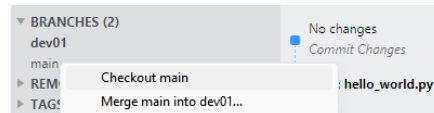
1. Switch back to the `main` branch:

Commandline

```
git checkout main
```

GUI

`main` ⇒ Right click ⇒ Checkout `main`



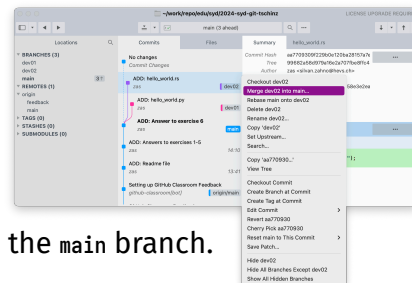
2. Merge the `dev02` branch into `main`:

Commandline

```
git merge dev02
```

GUI

Select Commit ⇒ Merge `dev02` into `main` ...



The work of the `dev02` branch is now integrated into the `main` branch.

7.4 Merge dev01

Similarly, it is now time to merge the `dev01` branch into `main`.

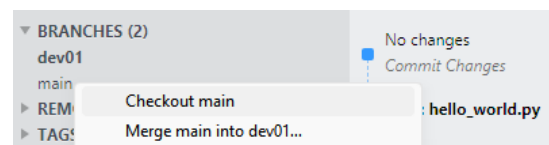
1. Ensure you are on the `main` branch:

Commandline

```
git checkout main
```

GUI

`main` ⇒ Right click ⇒ Checkout `main`



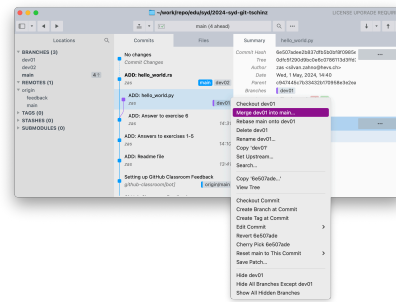
2. Merge the `dev01` branch into `main`:

Commandline

```
git merge dev01
```

GUI

Select Commit ⇒ Merge `dev01` into `main` ...



7.5 Final Result

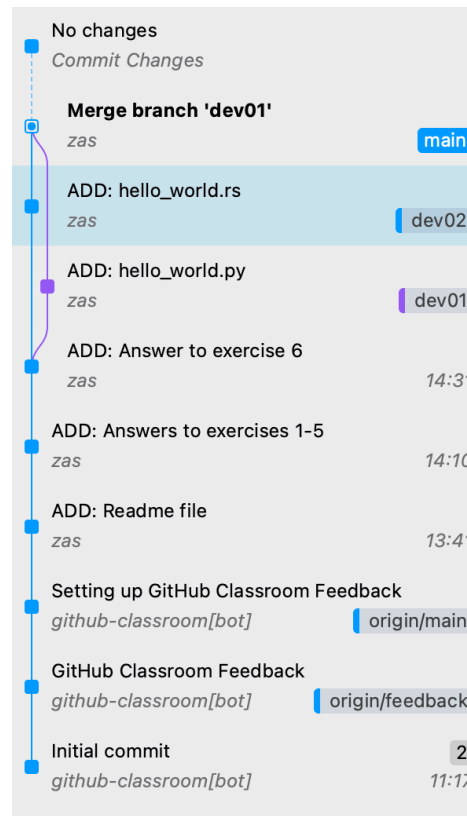


Figure 28 - Status after repo branches dev01 and dev02 have been merged



By default, the repository might only display a blue line. Since the work is very linear, Sublime Merge hides the commits. You can force the display of all commits by clicking on the button  on the commit lifeline.

1. Push your repository to the remote GitHub server. **Three push commands are required to push each branch separately:**

Commandline

GUI

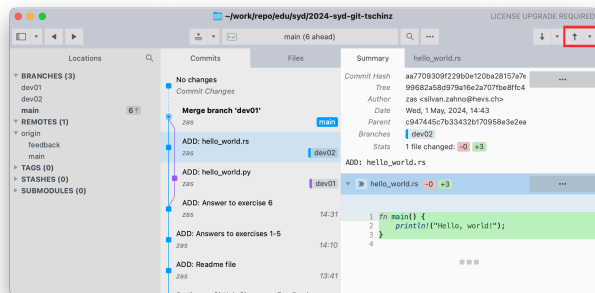
Checkout dev01 ⇒ Top Right Arrow ⇒ Push changes



```
git push origin main
git push origin dev01
git push origin dev02
```

Checkout dev02 ⇒ Top Right Arrow ⇒ Push changes

Checkout main ⇒ Top Right Arrow ⇒ Push changes



2. Tag the `main` branch with the tag `chapter3` as presented in chapter [Section 6.10](#). Don't forget to push the tag!



Check online on Github that the tag `chapter3` has been successfully created and pushed, along with your latest changes.



The status of your repository should be similar to [Figure 28](#) in additions to the tags `chapter2` and `chapter3`. This is part of the evaluation.



This chapter is to be completed in the same repository as before.



Make sure you have committed and pushed all your changes before starting this chapter, and that these have been tagged by chapter3.

8 | Gitgraph

The [Figure 30](#) shows an example of a git repository. Name all the elements that can be seen in the image (points 1-10).

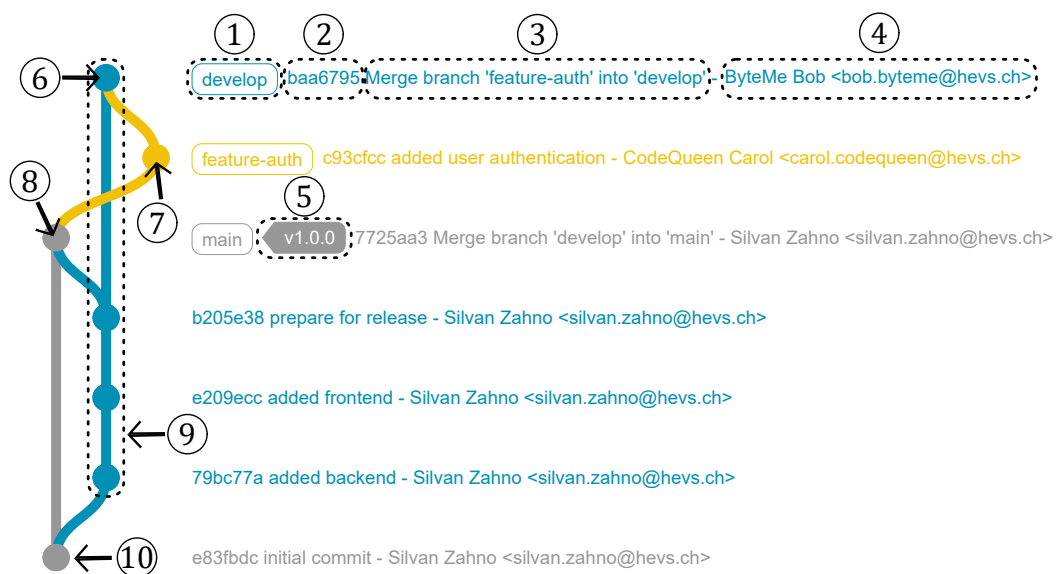


Figure 30 - Example of a Gitgraph

Task 7

1. Name all elements that can be seen in the image [Figure 30](#) (points 1-10)

2.



Commit your changes

3. Push your local repository to your cloud GitHub repository

4. Tag and push the tag chapter4 according to [Section 6.10](#)



Check online on Github that the tag `chapter4` has been successfully created and pushed, along with your latest changes.
Congratulations, the work on your first GitHub repository is finished.



This chapter is to be completed in a **different repository** than before.



Make sure you have committed all your changes before starting this chapter, and that these have been tagged by chapter4 and pushed online. Changes, commits, tags ... that are not available online will count as unfinished work for the grading of the lab. You can check online under [Github](#) if your repository is available and up to date.

9 | Gitflow

For this task, use the Gitflow philosophy presented in the course. You will all collaborate on the following Git repo as if you were forming a development team:

<https://github.com/hei-synd-syd/2026-syd-gitflow> [3]

This is a public Git repo hosted on Github.

9.1 Fork

For security reasons, you are not allowed to work directly on this repository. You need to create your own copy (fork) in order to make changes. Therefore, please create a fork of this repository in your GitHub account. To do this, use the fork button in the Github web interface.

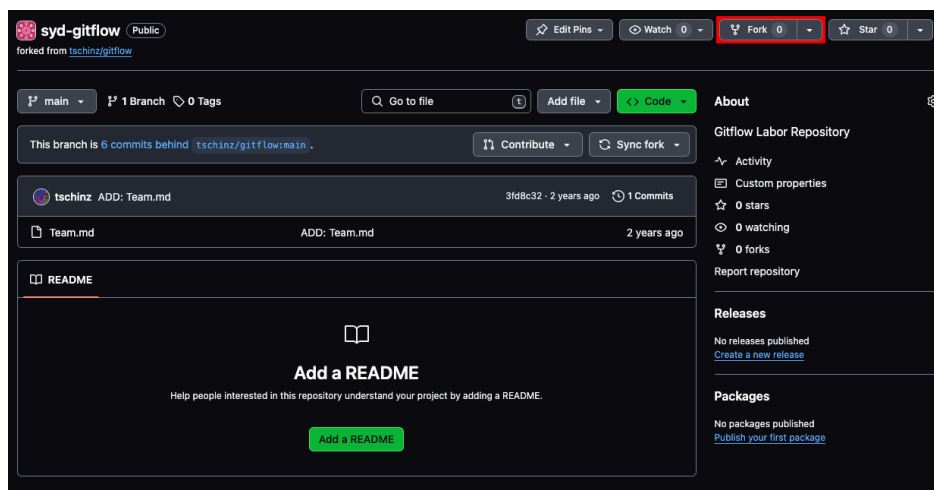


Figure 31 - Fork button for a GitHub repository

Then clone this forked repo. The URL of your new repo will look like this:

```
git clone https://github.com/<username>/<year>-syd-gitflow.git
```



9.2 Parallel collaboration



You will all edit the same file. To avoid conflicts, edit **only the line that is relevant to you**.

Ask the professor or teaching assistant to know your line number!



Edit the `Team.md` file.

Answer the question based on the number you received.



Commit and push your branch to your fork of the repository on GitHub.

9.3 Pull Request

Now that your changes are ready, it is still necessary to integrate them into the original repository <https://github.com/hei-synd-syd/2026-syd-gitflow>.

Since you do not have the rights to work on it directly, the only possibility is to make a merge request - pull request - on the original repository.

Thus, the administrator can accept the changes, ask you to correct items before accepting the merge, discuss issues, etc.

To do this, use the interface of the GitHub website ⇒ Contribute ⇒ Open pull request.

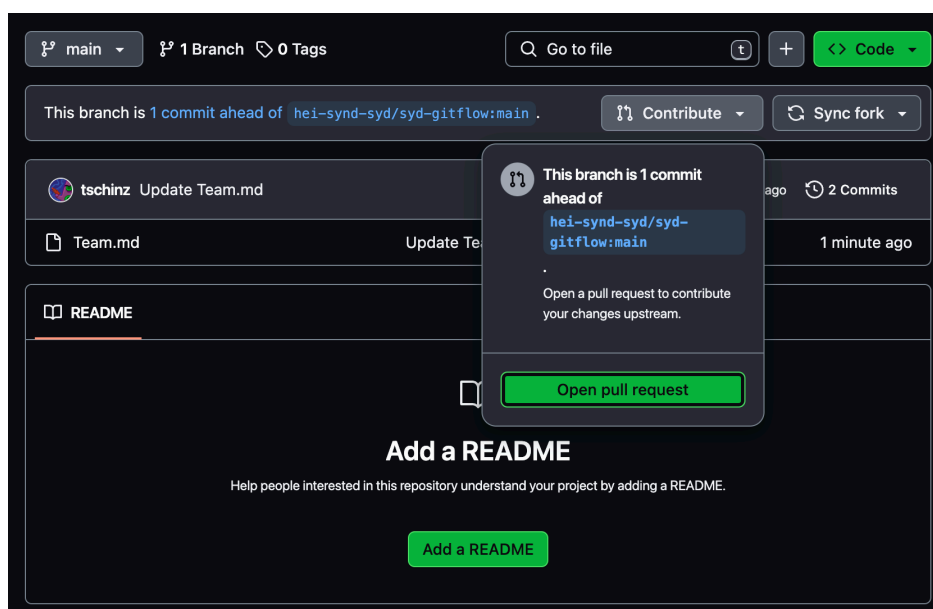


Figure 32 - Create Pull Request on Github

You will need to give a meaningful title and a short description to your pull request. It is possible to write the description in Markdown - [Section 6.6](#).



The incomplete, poorly described, poorly titled pull requests ... will be rejected.
When the pull requests are ready, the merges are done by the teacher and assistants.



Check online under <https://github.com/hei-synd-syd/2026-syd-gitflow/pulls> that:

- Your pull request has been correctly **opened**
- It contains a clear **title**
- It contains a clear **description**

You can still update your pull request description if necessary until it is accepted / rejected.



Missing pull requests will count as uncompleted work for the lab grading.



10 | Extras

This optional chapter can be started provided the previous tasks have been completed. There are 2 tasks to do:

1. Put your own project on Github and provide it with a README.md and a CI/CD.
2. Follow the “Learn Git Branching” website

10.1 Own project

- Put a current project you are working on, on github.
- Create a README.md file for the project using the [Markdown Syntax](#). The README.md should include the following:
 - Title
 - Image
 - Description of the project
 - Explanation how to run / use the project
 - List of authors
- Now create a github action to turn the README.md into a PDF on every push. For this find a suitable [github action](#) and add it to your project.



If you need help to create the github actions. Checkout the following [hint](#).

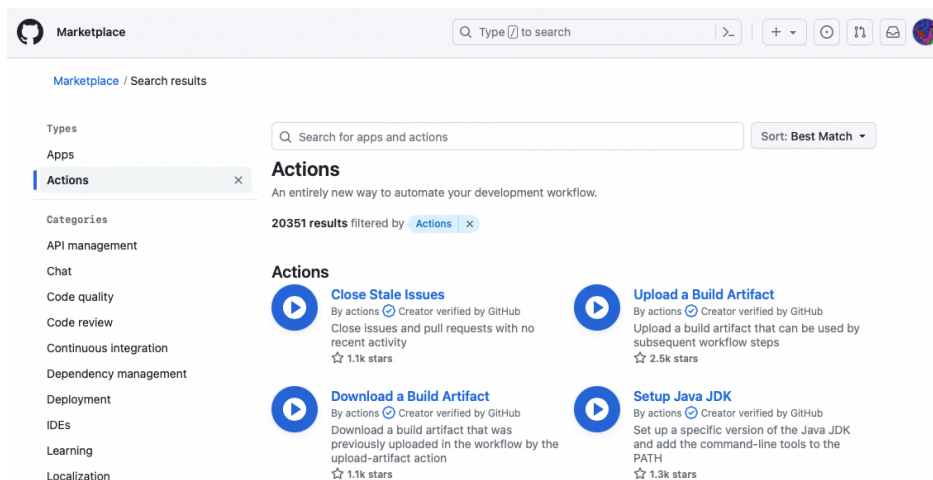


Figure 33 - Github Action Marketplace



10.2 Learn Git Branching

Follow the Tutorial on <https://learngitbranching.js.org>.

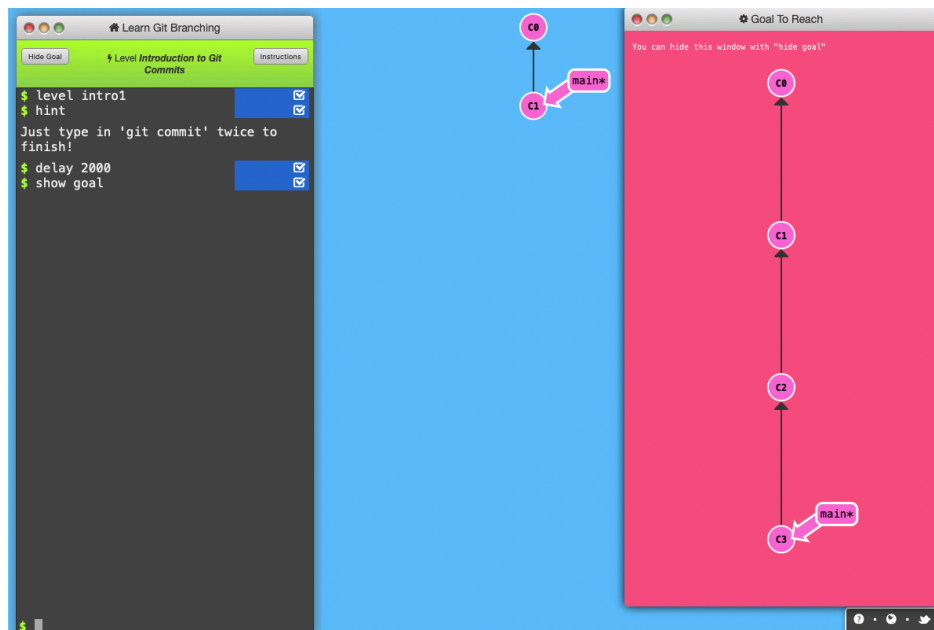


Figure 34 - Learn git branching website



Bibliography

- [1] T. Linus, "Git." Accessed: Apr. 25, 2023. [Online]. Available: <https://git-scm.com/>
- [2] "Sublime Merge - Git Client from the Makers of Sublime Text." Accessed: Apr. 25, 2023. [Online]. Available: <https://www.sublimemerge.com/>
- [3] tschinz, "Tschinz/Gitflow." Accessed: Apr. 25, 2023. [Online]. Available: <https://github.com/tschinz/gitflow>
- [4] gitlab, "Git Cheatsheet." 2023.
- [5] "GitHub Git Spickzettel." Accessed: Apr. 25, 2023. [Online]. Available: <https://training.github.com/downloads/de/github-git-cheat-sheet/>



11 | Appendix

A | GIT commands

[Github git cheatsheet \[4\], \[5\]](#)

AA Review changes and make a commit transaction.

```
git status
```

Lists all new or changed files ready for commit.

```
git diff
```

Displays file changes that have not yet been indexed.

```
git add [file]
```

Indexes the current state of the file for versioning.

```
git diff --staged
```

Shows the differences between the index ("staging area") and the current file version.

```
git reset [file]
```

Takes the file from the index, but preserves its contents.

```
git commit -m "[descriptive message]"
```

Adds all currently indexed files permanently to the version history.

AB Synchronize changes

Register an external repository (URL) and swap the repository history.

```
git fetch [remote]
```

Downloads the entire history of an external repository.

```
git merge [remote]/[branch]
```

Integrates the external branch with the current locally checked out branch.



```
git push [remote] [branch]
```

Pushes all commits on the local branch to GitHub.

```
git pull
```

Pulls the history from the external repository and integrates the changes.



B | Most used Git commands

BA Start a working area

- `clone` - Clone a repository into a new directory
- `init` - Create an empty Git repository or reinitialize an existing one

BB Work on the current change

- `add` - Add file contents to the index
- `mv` - Move or rename a file, a directory, or a symlink
- `reset` - Reset current HEAD to the specified state
- `rm` - Remove files from the working tree and from the index

BC Examine the history and state

- `log` - Show commit logs
- `show` - Show various types of objects
- `status` - Show the working tree status

BD Grow, mark and tweak your common history

- `branch` - List, create, or delete branches
- `checkout` - Switch branches or restore working tree files
- `commit` - Record changes to the repository
- `diff` - Show changes between commits, commit and working tree, etc
- `merge` - Join two or more development histories together
- `rebase` - Reapply commits on top of another base tip
- `tag` - Create, list, delete or verify a tag object signed with GPG

BE Collaborate

- `fetch` - Download objects and refs from another repository
- `pull` - Fetch from and integrate with another repository or a local branch
- `push` - Update remote refs along with associated objects